

Exercises from Ch. 7: Extension of Rings, Lang

Matthew Gergley

Last Updated: June 9, 2026

Disclaimer: The proofs given are completed by myself. Thus, they are also reviewed by myself to ensure no logic flaws and incorrect arguments/deductions/conclusions. If you find any error in my proofs, please do not hesitate to contact me so we can discuss any potential errors and I can update the document as needed.

Exercises

Exercise 0.1. (Exercise 2)

Let A be an entire ring and K its quotient field. Let t be transcendental over K . If A is integrally closed, show that $A[t]$ is integrally closed.

Proof. Assume that A is integrally closed. Hence, if $x \in K$ is integral over A , then $x \in A$, i.e. $\overline{A}^K = A$. Consider the quotient field of $A[t]$,

$$K(t) = \left\{ \frac{f(t)}{g(t)} : f(t), g(t) \in A[t], (f(t), g(t)) = 1 \right\}.$$

Let $\alpha(t) = \frac{f(t)}{g(t)} \in K(t)$ be integral over $A[t]$. Hence, there exists $b_{n-1}(t), \dots, b_0(t) \in A[t]$ such that

$$\alpha(t)^n + b_{n-1}(t)\alpha(t)^{n-1} + \dots + b_1(t)\alpha(t) + b_0(t) = 0.$$

Multiplying through by $g(t)^n$, we obtain

$$h(t) = f(t)^n + b_{n-1}(t)f(t)^{n-1}g(t) + \dots + b_0(t)g(t)^n = 0.$$

Observe that since the coefficients of $b_i(t)$ are in $A \subseteq K$, $h(t) \in K[t] \cong K[X]$ (by the transcendence of t over K). Rearranging the above equation,

$$f(t)^n = -g(t)(b_{n-1}(t)f(t)^{n-1} + \dots + b_0(t)g(t)^{n-1}),$$

so $g(t) \mid f(t)^n$. But by the definition of $\alpha(t) \in K(t)$, $(f(t), g(t)) = 1$, and the fact that $K[t] \cong K[X]$ is a PID (and thus a UFD), we must have that $g(t) \in K^\times$. Hence, $\alpha(t) = \sum_{i=0}^m d_i t^i \in K[t] \cong K[X]$ with $d_i \in K$. Substituting this into our integral expression at the start, we obtain

$$H(t) = \alpha(t)^n + b_{n-1}(t)\alpha(t)^{n-1} + \dots + b_0(t) = 0 \in K[t].$$

Since t is transcendental over K , if $H(t) \in K[t]$ equals 0, then $H(t)$ must be the zero polynomial. Thus, expanding and collecting powers of t , the coefficients of t must be zero which gives $d_i \in K$ is integral over A for all $i = 0, \dots, m$. By the assumption of A being integrally closed in K , we have $d_i \in A$ and thus $\alpha(t) = \sum_{i=0}^m d_i t^i \in A[t]$. Therefore, $A[t]$ is integrally closed. □

For the following exercises, you can use §1 of Chapter X.

Exercise 0.2. (Exercise 3)

Let A be an entire ring, integrally closed in its quotient field K . Let L be a finite separable extensions of K , and let B be the integral closure of A in L . If A is Noetherian, show that B is a finite A -module. [Hint: Let $\{\omega_1, \dots, \omega_n\}$ be a basis of L over K . Multiplying all elements of this basis by a suitable element of A , we may assume without loss of generality that all ω_i are integral over A . Let $\{\omega'_1, \dots, \omega'_n\}$ be the dual basis relative to the trace, so that $\text{Tr}(\omega_i \omega'_j) = \delta_{ij}$. Write an element α of L integral over A in the form

$$\alpha = b_1 \omega'_1 + \dots + b_n \omega'_n$$

with $b_j \in K$. Taking the trace $\text{Tr}(\alpha \omega_i)$ for $i = 1, \dots, n$, conclude that B is contained in the finite module $A\omega'_1 + \dots + A\omega'_n$. Hence B is Noetherian.

Proof. We follow the hint. Let $\{\omega_1, \dots, \omega_n\}$ be a basis for L over K . Since L/K is finite, by V Prop. 1.1, L is an algebraic extension over K . Thus, for all $i = 1, \dots, n$ ω_i is algebraic over K . Consider the irreducible polynomials

$$\text{Irr}(\omega_i, K, X) = f_i(X) = X^{m_i} + c_{m_i-1}X^{m_i-1} + \dots + c_0.$$

Observe that these are monic via the multiplication of an element in K . Since the coefficients are in K , the denominators are in A . For all i , choose some nonzero $d_i \in A$ that clears the denominator of the coefficients (i.e. the least common multiple of the denominators). Hence, $d_i f_i(X) \in A[X]$. Consider $y_i = d_i \omega_i$. Thus,

$$f_i(\omega_i) = f_i\left(\frac{y_i}{d_i}\right) = \left(\frac{y_i}{d_i}\right)^{m_i} + c_{m_i-1} \left(\frac{y_i}{d_i}\right)^{m_i-1} + \dots + c_0 = 0.$$

Multiplying by $d_i^{m_i}$,

$$y_i^{m_i} + d_i c_{m_i-1} y_i^{m_i-1} + \dots + d_i^{m_i} c_0 = 0.$$

Observe that $d_i^{m_i-j} c_j = d_i^{m_i-j-1} (d_i c_j) \in A$ since $d_i \in A$, $c_j \in A$, and $d_i^{m_i-j-1} \in A$. Therefore, $y_i = d_i \omega_i$ satisfies the monic

$$Y^{m_i} + d_i c_{m_i-1} Y^{m_i-1} + \dots + d_i^{m_i} c_0 \in A[Y],$$

so $d_i \omega_i$ is integral over A . Thus, if we take the product of powers of d_i for all i , call it $d \in K^\times$, we see that $d\omega_i$ is also integral over A for all i (via the divisibility arising from the product). Hence, $\{d\omega_1, \dots, d\omega_n\}$ is a basis with all elements integral over A . Thus, without loss of generality, we may assume that $\{\omega_1, \dots, \omega_n\}$ is a basis for L/K with all elements integral, since if not, we can construct a basis by the above argument. By VI Corollary 5.3, there exists another basis $\{\omega'_1, \dots, \omega'_n\}$ for L/K such that $\text{Tr}_K^L(\omega_i \omega'_j) = \delta_{ij}$. Let $\alpha \in B$. Hence, $\alpha \in L$ and α is integral over A . Using the dual basis relative to the trace, write

$$\alpha = \sum_{i=1}^n b_i \omega'_i, \quad b_i \in K.$$

Taking the trace of $\alpha \omega_i$ for each i , we have

$$\begin{aligned} \text{Tr}_K^L(\alpha \omega_i) &= \text{Tr}_K^L\left(\sum_{j=1}^n b_j \omega'_j \omega_i\right) \\ &= \sum_{j=1}^n b_j \text{Tr}_K^L(\omega'_j \omega_i) \\ &= \sum_{j=1}^n b_j \delta_{ij} = b_i, \quad \forall i = 1, \dots, n. \end{aligned}$$

Since α is integral over A and ω_i are integral, $\text{Tr}_K^L(\alpha \omega_i) \in K$ is integral over A . By the definition of A integrally closed, $b_i = \text{Tr}_K^L(\alpha \omega_i) \in A$. Since this holds for all i ,

$$\alpha = \sum_{i=1}^n b_i \omega_i = A\omega_1 + \cdots + A\omega_n.$$

Hence, since $\alpha \in B$ was arbitrary,

$$B \subseteq A\omega_1 + \cdots + A\omega_n.$$

Clearly, $A\omega_1 + \cdots + A\omega_n$ is a finitely-generated A -module and since A is Noetherian, by X Prop. 1.4, $A\omega_1 + \cdots + A\omega_n$ is Noetherian. And finally, since B is a submodule of $A\omega_1 + \cdots + A\omega_n$, via X Prop. 1.1 and definition of being Noetherian, B is finitely-generated and is Noetherian.

□